

Multimodal Policy Design Assignment

Last week we explored how multimodal sensing can help our robotic cellist. This week, study how to design a control policy that turns audio, force/touch, and robot motion into useful actions.

DUE: TUESDAY, SEPTEMBER 29, 2026

Finish and share slides before the meeting.

Present at Tuesday's meeting.

Your task

Choose **any 1-2 papers** on page 2. Work **individually or in a team**.

Prepare **one 3-slide deck** and a **2-3-minute presentation** per individual/team. All team members must speak. Answer the question for each selected paper within the same deck.

Presentation structure

SLIDE	WHAT TO COVER	TIME
1. Policy	Sketch the inputs, fusion, policy, and actions. State the training signal and which observations are available during execution.	40 sec
2. Evidence	Use one figure or table to support your answer. Explain the comparison, metric, and result. Cite the paper and figure/table number.	60 sec
3. Project design	Propose a cello policy with input history, fusion, action output, and update timing. Name one test and a success metric.	40 sec

Connect the paper to our robotic cellist

Use a sustained bow stroke as a concrete example. Consider recent audio, contact force, and robot state, with vision if useful. Define actions that our hardware can command, such as bow-motion commands or force targets if supported.

Label proposed cello adaptations clearly. Support design choices with the paper's method and evidence, and identify one uncertainty to test. A test could compare the proposed policy with a version that omits one sensor, using a defined sound-quality or contact-stability metric.

Upload your slides to the **Week 6 folder** in our [shared Google Drive](#) before the meeting.

In this assignment, multimodal refers to multiple sensor types. A multimodal action distribution describes several possible actions; it is a separate concept.

Choose 1-2 papers

Click a title to open the paper. Each paper has one guiding question.

Use its method and evidence to justify your answer and proposed cello adaptation.

1. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

Action chunks and feedback timing | arxiv.org/abs/2303.04137

Question: How would you condition a Diffusion Policy on audio, contact force, and robot state to generate bowing actions while choosing an execution horizon that still allows timely feedback?

2. ManiWAV: Learning Robot Manipulation from In-the-Wild Audio-Visual Data

Audio fusion and temporal context | Returning paper | arxiv.org/abs/2406.19464

Question: How should a cello policy encode and synchronize recent audio with robot state so that bowing commands respond to sound changes despite sensor delay and motor noise?

3. FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning

Training a policy to use force | Returning paper | arxiv.org/abs/2502.17432

Question: What training curriculum would you design, inspired by FACTR, to prevent our cello policy from ignoring force feedback?

4. Reactive Diffusion Policy: Slow-Fast Visual-Tactile Policy Learning for Contact-Rich Manipulation

Fast feedback during action execution | arxiv.org/abs/2503.02881

Question: How would you divide bowing control between a slow policy and a fast feedback policy so that a sudden contact-force change can be corrected during an action chunk?

5. ViTacFormer: Learning Cross-Modal Representation for Visuo-Tactile Dexterous Manipulation

Cross-modal fusion and future contact prediction | arxiv.org/abs/2506.15953v2

Question: How could predicting future contact signals help our cello policy choose its next bowing action, and how would you test the value of that prediction?

6. ForceMimic: Force-Centric Imitation Learning with Force-Motion Capture System for Contact-Rich Manipulation

Force targets and the policy-to-controller interface | arxiv.org/abs/2410.07554

Question: Which bowing variables should our policy output as motion targets and which as force targets, using ForceMimic's policy-to-controller interface as a guide?